

68513 Plasmonics Computer Lab

You can choose either LSP or SPP – you will need to refer to lecture notes for background.

LSP

The system being simulated is a metal shell, illuminated from $-z \rightarrow +z$, Ex polarization. It uses Mie theory which matches spherical harmonics at the interfaces.

1. Run the example file for a shell to see what you get.

Make sure you understand what is being shown. You should get figures with a spectrum, angular scattering chart & near-field plots. The second figure gives more detail on each of the peaks in the first figure. The angular scattering angle is measured from the forward direction (0=forward, 180=back). The field plots show $|E|$ with power flow as red lines.

2. Interpret the results.

What's the difference between the two peaks (e.g. absorption vs scattering, E fields & power flow)? Which modes are they (dipole or higher)?

3. Explore the effect of parameters:

[recommended fixed parameters are core & background permittivity 1, outer radius 10nm]

- geometry of particle (e.g. varying shell thickness)
- choice of metal (try at least 2)

If you wish, you may *additionally* investigate the effect of the core index.

How do these parameters affect peak position, strength and character?

How can the results be related to physical models (e.g. based on charge)?

SPP

The system being simulated is a metal film, with both Kretschman & Otto prism configurations. It uses planewaves & thin film equations to match fields at the interfaces. Important: the angle shown is the internal angle, i.e. inside the prism.

1. Run the example file to see what you get: make sure you understand what is being shown.

You should get a figure with energy-momentum maps, angular scan & near-fields.

The near-field plots show the normal component of electric field – this is related to the equivalent surface charge. The prism is on the left. The white lines are the electric field direction, and the black lines power flow.

2. Interpret the results. How does the angular scan relate to the map?

What are the dips showing?

3. Explore the effect of parameters:

[recommended fixed parameters as per example]

- film thickness
- choice of metal (try at least 2)

If you wish, you may *additionally* investigate the effect spacer thickness (Otto configuration).

How does these parameters affect SPP position, depth, width, and charge distribution?

How can the results be related to physical models (e.g. based on charge)?